

PROJECT REPORT

On

“Cafe Inventory System”

Submitted in the partial fulfillment of Bachelor of Computer
Applications

To

Uttaranchal University



UTTARANCHAL
UNIVERSITY
DEHRADUN

NAAC
GRADE A+



Uttaranchal School of Computing Sciences
SESSION- (2025-2026)

Under the Supervision of

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B.C.A. 2nd Semester

Section- F

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The most awaited moment of any endeavor is successful completion, but nothing can be done successfully if done alone. Success is the outcome of contribution and consistent help of various person and we thank those ones who helped us in successful completion of this project.

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DECLARATION

I hereby declared that the project report entitled “**Cafe Inventory System**” submitted by **Komal** to **Uttaranchal School of Computing Science**. The project report was done under the guidance of Ms. **Manisha Khanduja, Assistance Professor-USCS**. I further declare that the work reported in this project report has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this university or any other university or institute.

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CERTIFICATE

This is to certify **Komal** student of BCA 2nd Semester, of **Uttaranchal School of Computing Sciences, Dehradun**, has completed the Project Report for the topic “**Cafe Inventory System**” for the session (2025- 2026).

Under the guidance of:

Ms. Manisha Khanduja

Assistant professor- USCS

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TABLE OF CONTENTS

S.NO.	CONTENT	PAGE NO.
	Acknowledgement	I
	Declaration	II
	Certificate	III
1.	Introduction	01-04
	1.1 Objective of the project	03
	1.2 Scope of the project	04
2.	System Analysis/Requirement	05-9
	2.1 System analysis	06
	2.2 System requirement	07
	2.3 Feasibility Study	08
	2.4 Economic Feasibility	09
3.	Planning	10-16
	3.1 Project Planning Overview	11
	3.2 Gantt Chart	12
	3.3 Flow Chart	13
	3.4 Data Flow Diagram(DFD)	14-15
4.	System Design And Module Structure	16-19
	4.1 Number of Modules and Description	17
	4.2 Data Structures Used in the Project	18
	4.3 Proposed Security Mechanisms	19
5.	Testing	20
	5.1 Unit Testing	
	5.2 Integrating Testing	
7.	Future Scope and Further Enhancement	21-22
8.	Code	23-34
9.	Bibliography	35

Introduction

The Cafe Inventory System is a console-based project developed to manage and maintain the stock details of a cafe in an efficient and organized manner. Inventory plays a very important role in the smooth functioning of any food business because raw materials like coffee beans, milk, sugar, vegetables, snacks, and other items are used daily. In small cafes, inventory is usually maintained manually using notebooks or simple spreadsheets, which consumes time and often leads to errors. Problems such as incorrect stock calculation, missing records, difficulty in searching past entries, and lack of proper monitoring can affect the overall management of the cafe. There is also no automatic alert system to notify when stock is low, which may result in shortages during business hours.

To overcome these limitations, the proposed Cafe Inventory System aims to computerize the inventory management process using the C++ programming language. This project is a console-based, menu-driven application developed with basic programming concepts such as functions, loops, conditional statements, arrays, structures or classes, and file handling for permanent data storage. The system allows the user to add new items, update stock quantities, delete existing records, and view available inventory easily. It automatically calculates the remaining stock after each transaction and securely stores data for future reference. By implementing this system, manual workload is reduced, accuracy is improved, records are better organized, and decision-making regarding stock purchasing becomes easier.

In addition, the system provides a simple interface that allows the user to manage inventory operations without requiring advanced technical knowledge. The application helps maintain proper records of all items and ensures that stock information is updated whenever changes occur. This makes it easier for cafe owners or managers to track available resources and plan purchases in advance. The system also improves efficiency by reducing paperwork and saving time in daily inventory management tasks.

Furthermore, the Cafe Inventory System can help in maintaining better control over stock usage and preventing wastage of resources. By keeping a clear record of available quantities, the cafe management can monitor which items are used frequently and which items need to be restocked. This information is useful for improving inventory planning and ensuring that important ingredients are always available when required.

Overall, the Cafe Inventory System provides a simple, reliable, and effective solution for managing cafe inventory in a systematic and computerized way. It helps reduce human errors, improves record keeping, and increases the efficiency of inventory management. This project also demonstrates the practical application of programming concepts learned in C++, making it a useful learning experience for students while solving a real-world problem.

1.1 Objective of the Project

1. **Record Maintenance:** The system helps in keeping all inventory data well-organized and properly structured. This eliminates confusion caused by manual record-keeping methods like notebooks or scattered files.
2. **Permanent Data Storage:** The system uses file handling techniques to store inventory data permanently. This ensures that all records are saved even after the program is closed or the system is restarted. It allows the cafe owner to maintain long-term records of stock, which can be useful for analysis, planning, and future decision-making.
3. **Error Reduction:** Manual inventory management often leads to errors such as incorrect calculations, duplicate entries, or missing records. The Cafe Inventory System reduces these errors by automating various operations like adding, updating, and deleting items.
4. **Restock Management:** The system helps in identifying low-stock items so that they can be restocked on time. By regularly monitoring inventory levels, the cafe owner can ensure that essential items like coffee, milk, or snacks do not run out during business hours. This improves customer satisfaction and avoids interruptions in service.

1.2 Scope of the project

- **Console-Based Development:** The project will be developed as a console-based menu driven application using the C++ programming language.
- **Menu-Driven Functionality:** The system will contain different options such as Add Item, Update Item, Delete Item, Search Item, and View All Items to manage inventory efficiently.
- **Data Structure Implementation:** Each item record will be stored using structures or classes within the program to organize data properly.
- **File Handling for Data Storage:** File handling techniques will be used to store data permanently so that records remain saved even after closing the program.
- **Automatic Stock Management:** The program will automatically calculate stock quantity whenever inventory changes occur.
- **Target User Environment:** The system will be designed mainly for small cafes where a single user (owner or staff member) manages the inventory.
- **User-Friendly Interface:** The interface will be simple and easy to understand so that beginners can operate the software without technical knowledge.

System Analysis/Requirement

System Analysis serves as the critical foundation of this project, involving a detailed study of the existing inventory management practices in small cafes to identify inefficiencies and design a better computerized solution. The main focus of this phase is to examine the limitations of the current manual stock management system, which often suffers from inaccuracy, time consumption, data duplication, and a high risk of human errors. In a manual setup, every stock transaction—whether new ingredients are purchased, items are consumed for preparation, or expired products are discarded—must be recorded by hand in registers or notebooks. This process is labor-intensive and makes tracking stock levels a difficult task. For example, checking the availability of a particular ingredient or calculating the remaining stock at the end of the day requires manually reviewing multiple entries, which leads to delays and mismanagement.

The proposed Cafe Inventory System addresses these issues by introducing a structured and automated digital solution. Instead of maintaining multiple physical records, all inventory data will be stored in a centralized system using file handling techniques in C++. Each item will have specific details such as item name, quantity, and price, ensuring that information is recorded systematically and consistently. The system allows quick searching, updating, and displaying of stock records, reducing the time required to retrieve information. It also performs automatic stock calculations whenever items are added or consumed, eliminating manual counting errors and ensuring accuracy.

Additionally, the system incorporates basic logical operations to support inventory control, such as identifying low-stock items and maintaining updated quantity records after each transaction. Operationally, it is simple to use and designed for small cafes where a single user manages stock. Economically, it is cost-effective because it replaces paper-based systems without requiring expensive software. Overall, the system analysis confirms that the Cafe Inventory System is a practical, reliable, and efficient solution for improving inventory management in small-scale cafe businesses.

2.1 System analysis

System analysis is the process of examining the current system and identifying its limitations. In the traditional cafe inventory system, stock records are maintained manually using registers or notebooks. This method becomes inefficient as the number of items and daily transactions increases.

The following issues were identified during system analysis:

- Manual record keeping is time-consuming.
- Searching for specific stock items takes more time.
- High possibility of human errors in calculations and entries.
- Difficulty in maintaining accurate and updated stock records.
- No automated tracking of stock usage and restocking.

Lack of security: Physical records are susceptible to damage, loss, or unauthorized changes, with no backup available if registers are misplaced.

Based on these observations, it was concluded that a computerized Cafe Inventory System is required to improve efficiency, accuracy, and proper stock management.

2.2 System requirement

Hardware Requirements

- Laptop
- Minimum 4 GB RAM
- Basic system capable of running C++ programs

Software Requirements

- Windows operating system
- C++ compiler (Dev C++)
- Console support

Functional Requirements

- Update stock quantity
- Delete inventory items
- Search for specific items
- Display complete inventory list
- Automatic stock calculation
- File handling for data storage

Non-Functional Requirements

1. User-friendly interface
2. Fast response time
3. Accurate inventory records
4. Reliability of the system
5. Easy maintenance

2.3 Feasibility Study

A feasibility study is conducted to determine whether the proposed system is practical and achievable. It evaluates different aspects such as technical, operational, and economic feasibility.

2.3.1 Technical Feasibility

Technical feasibility examines whether the required technology and resources are available to develop the system. The proposed Cafe Inventory System is developed using C++, which is a standard programming language taught in the BCA course.

The system requires:

- A computer with basic configuration.
- A C++ compiler such as Code Blocks or Visual Studio.
- Basic knowledge of programming concepts.

Since these resources are readily available, the project is technically feasible.

2.3.2 Operational Feasibility

Operational feasibility evaluates whether the system will function effectively in the given environment and whether users can operate it easily.

The proposed system is:

- Simple and user-friendly.
- Menu-driven and easy to navigate.

2.4 Economic Feasibility

- The project does not require expensive hardware or advanced equipment.
- It runs on a standard computer or laptop.
- No additional external devices are required.
- The system is developed using C++ which is freely available.
- No paid software or licensed tools are needed.
- Development cost is minimal as it is an academic project.
- Maintenance cost is very low.
- No internet or server cost is required.
- Reduces manual labor cost by automating the inventory management process.
- Saves time, which increases overall efficiency.
- Suitable for small cafes with limited budget.

Planning

Project planning is an essential phase in software development. It involves defining project activities, estimating time required for each task, and organizing them in a structured manner. Proper planning ensures that the project is completed within the specified time frame and resources. It also helps in identifying dependencies between different tasks and managing the workflow efficiently.

For the Cafe Inventory System, project planning includes requirement analysis, system design, coding, testing, and documentation. These activities are organized systematically using project scheduling techniques such as Gantt charts.

The planning phase of the Cafe Inventory System is a crucial step before actual development begins, as it provides a clear direction and structure for the project. This phase focuses on analyzing the current inventory management process in small cafes and planning how the computerized system will replace manual record-keeping methods. Proper planning ensures that the system is not only functional but also practical and efficient in a real-world cafe environment. During this stage, the project lifecycle is clearly defined using basic System Development Life Cycle (SDLC) principles. The process begins with requirement analysis, followed by system design, coding, testing, and final implementation. Since this is an academic project, a simple and structured development approach is adopted. The planning also involves selecting appropriate tools, such as the C++ programming language and file handling techniques, to ensure smooth data storage and retrieval. The system is designed as a console-based application with a menu-driven interface for easy user interaction.

Risk assessment is also considered during planning. Possible challenges include incorrect data entry, loss of records during file handling, or logical errors in stock calculation. To minimize these risks, proper testing and debugging are planned at every stage of development. Time management is another important aspect of this phase, where the project is divided into small milestones such as requirement gathering, flowchart creation, coding modules, testing, and documentation. This structured planning helps prevent confusion, ensures timely completion, and maintains project quality.

3.1 Project Planning Overview

The Project Planning Overview provides a clear roadmap for the development of the Cafe Inventory System from initial concept to final submission. It ensures that all functional and non-functional requirements are addressed properly.

The first milestone is **Requirement Analysis**, where the necessary features such as adding items, updating stock, deleting records, searching items, and displaying inventory are identified. After this, the **System Design Phase** begins, which includes preparing flowcharts, algorithms, and basic data structure planning using structures or classes. This step acts as a blueprint for coding and ensures proper organization of the program.

Next, is the **Development and Coding Phase**, where the system is built using modular programming. Each function, such as Add Item, Update Item, Delete Item, and Display Item, is developed separately to make the program easy to understand and maintain. File handling is integrated carefully to ensure permanent data storage.

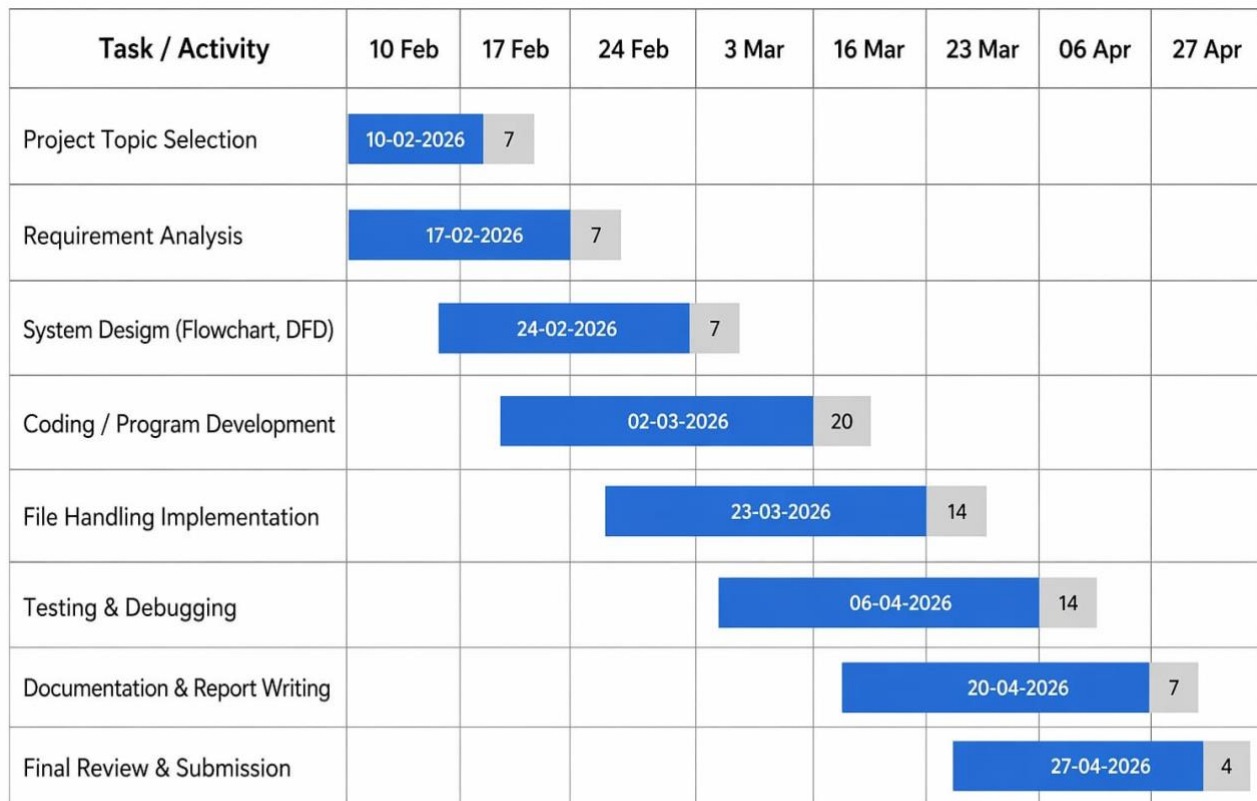
The **Testing Phase** follows, where the program is checked for logical errors, incorrect outputs, and file handling issues. Different test cases are applied to verify that stock updates and calculations work correctly. After successful testing, the project moves to the **Implementation and Documentation Phase**, where the final program is compiled, demonstrated, and documented properly in the project report.

This complete planning overview ensures that the Cafe Inventory System is simple, reliable, cost-effective, and suitable for small cafes. It acts as a structured guide that helps in developing an efficient inventory management solution.

3.2 Gantt Chart

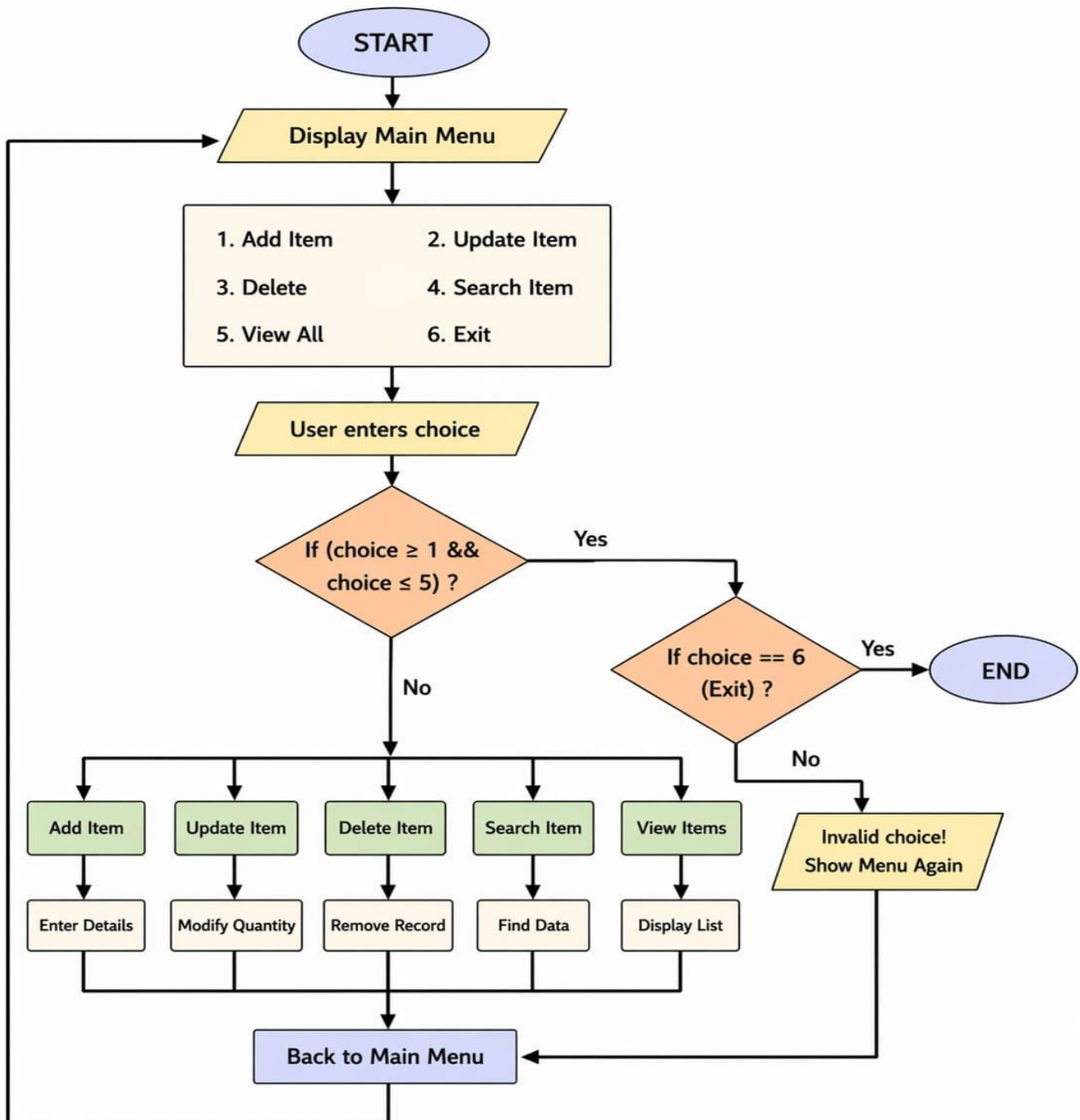
A Gantt Chart is a graphical tool used in project management to represent the schedule of tasks over a specific period of time. It shows different activities of a project along with their start time, duration, and completion in the form of horizontal bars.

In the Cafe Inventory System project, the Gantt Chart helps in planning and tracking tasks such as requirement gathering, system design, coding, testing, and implementation, ensuring the project is completed on time.



■ start date ■ duration(in days)

3.3 Flow Chart



3.4 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) represents how data moves within the system and how different processes interact with each other. It helps in understanding the flow of information between users, processes, and data storage in the Cafe Inventory System. The DFD is divided into different levels to show both the overall system and the internal processes.

1. Level 0 DFD (Context Diagram)

The Level 0 DFD, also known as the Context Diagram, represents the entire Cafe Inventory System as a single process and shows its interaction with external entities.

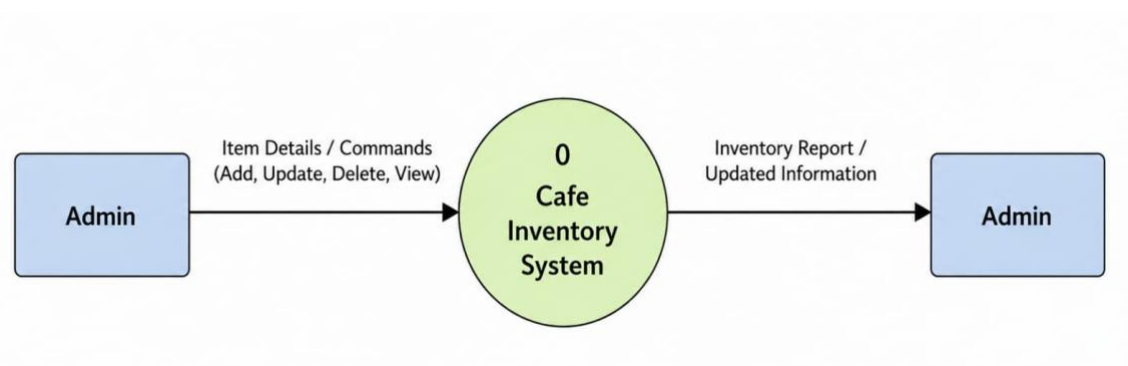
In the Cafe Inventory System, the Level 0 DFD consists of:

One main process: 0 Cafe Inventory System

One external entity: Admin

Data flow: Admin sends item details/commands (add, update, delete, view)

System returns inventory reports/updated information.



2. Level 1 DFD

The Level 1 DFD provides a more detailed view of the internal processes of the Cafe Inventory System. The main system is divided into multiple processes:

Add Item (1.0)

This process allows the admin to enter new item details such as ID, name, and quantity. The data is then stored in the inventory database for future use.

Update Item (2.0)

This process is used to modify existing item details in the inventory. The system updates the selected item information in the database.

Delete Item (3.0)

This process removes an item from the inventory based on its ID. The selected record is deleted from the database permanently.

View Inventory (4.0)

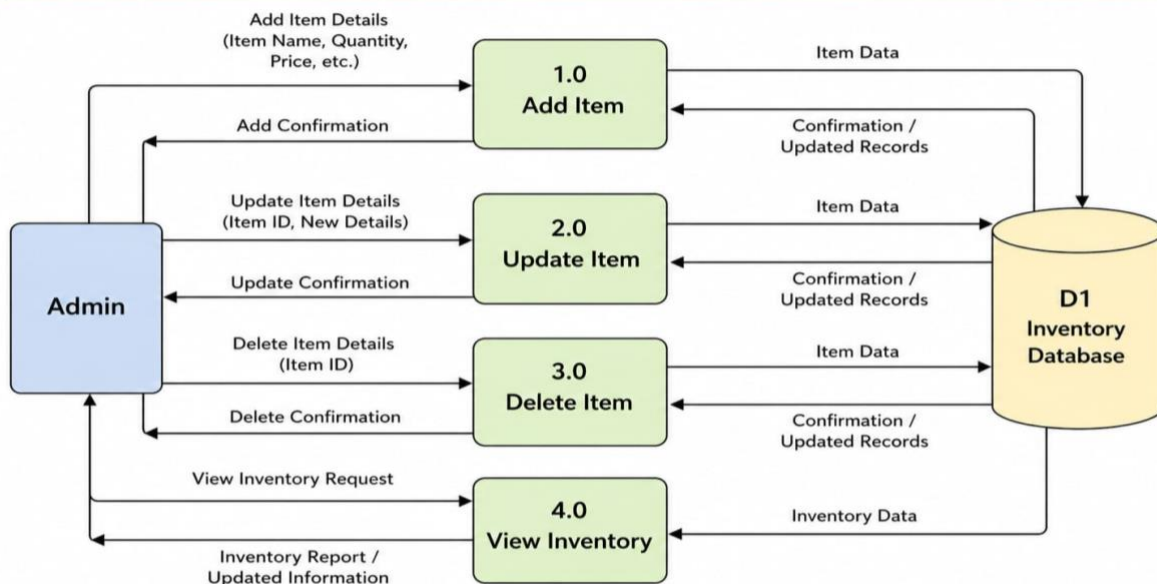
This process displays all available items in the inventory. It retrieves data from the database and shows it to the admin.

External Entity – Admin

The Admin is the user who interacts with the system to manage inventory operations. They provide input (add, update, delete, view) and receive output from the system.

Data Store – Inventory Database/File

The data store keeps all inventory records such as item ID, name, and quantity. It is used by processes to store, retrieve, update, and delete data.



System Design and Module Structure

The System Design phase acts as the bridge between system analysis and actual coding of the Cafe Inventory System. It defines the overall structure, modules, and data organization required to meet the specified requirements. The purpose of this stage is to create a clear blueprint so that the system is simple, efficient, and easy to maintain.

Since this is a console-based mini project developed in C++, the design follows a modular approach. The system is divided into separate modules such as item management, stock updating, and record display. Each module performs a specific task independently. This separation ensures that if one module needs modification, it does not affect the entire system. The architecture mainly includes: Presentation Layer – Console-based user interface (menu-driven system). Application Layer – Business logic for adding, updating, deleting and calculating stock. Data Layer – File handling system used for storing inventory records permanently.

The design also focuses on proper data organization. Each inventory item contains fields such as Item ID, Item Name, Quantity, and Price. Logical operations ensure that stock is automatically updated whenever changes are made. Basic validation checks are included to reduce incorrect data entry.

4.1 Number of modules and Description

The Cafe Inventory System is divided into four main modules. The **Item Management Module** handles adding new items, modifying item details, and deleting discontinued items from inventory. The **Stock Update Module** manages stock changes when items are purchased (restocked) or consumed. Automatically updates quantity. The **Search and Display Module** allows users to search for specific items and display the complete inventory list. Finally, the **File Handling Module** responsible for saving and retrieving data from files to ensure permanent storage of records.

4.2 Data Structure Used In Project

Data structures are used in a program to store and manage inventory data efficiently. In the Cafe Inventory System, different data structures are used to handle item data. These data structures help in organizing the program and performing operations.

➤ **Structure for Item Data**

A structure named Item is used to store details of each inventory item. It includes fields such as:

Item ID

Item Name

Quantity

Price

This structure helps in organizing item information systematically.

➤ **Array for Storing Items**

An array is used to store multiple item records. This allows easy access, updating, and searching of items within the program.

➤ **File Handling for Permanent Storage**

File handling is used to store item records permanently. Data is written to and read from a file to maintain records even after the program is closed.

4.3 Proposed Security Mechanisms

The security architecture of the Cafe Inventory System is designed to ensure the confidentiality, integrity, and availability of inventory data. The system protects important information such as stock records and item details from unauthorized access or accidental loss, ensuring reliable inventory management.

- **Role-Based Access Control (RBAC):** The system uses role-based access to control user activities. The Administrator (Cafe Manager) can add, update, and delete inventory items, while other staff members may only view the records. This prevents unauthorized modifications.
- **Secure Login and Authentication:** A login system with username and password is used to allow only authorized users to access the inventory management features.
- **Input Validation and Error Handling:** The program checks all user inputs to prevent invalid entries such as negative quantities or incorrect menu options, ensuring accurate and consistent data.
- **Activity Logging:** Important actions like adding, updating, or deleting inventory items are recorded to monitor system activities and track changes.
- **Data Backup and Storage:** Inventory data is stored using file handling so that records remain saved even after the program is closed, helping prevent data loss.

TESTING

Testing is performed to ensure that the Cafe Inventory System works correctly and meets the required functionality. It helps in identifying errors and verifying that each module performs as expected.

5.1 Unit Testing

- Each module was tested individually:
- Add Item module tested for correct data entry.
- Update module tested for accurate stock changes.
- Delete module tested for proper record removal.
- Search module tested for correct item retrieval.

5.2 Integration Testing

- After unit testing, all modules were combined and tested together. It was verified that:
- Stock updates correctly reflect in the inventory list.
- File records are saved and retrieved properly.
- The system handles different input conditions reliably.

SCOPE AND FUTURE ENHANCEMENTS

The proposed Cafe Inventory System is a console-based application developed using C++. Although the system successfully automates essential inventory management functions, there is significant scope for further improvement and enhancement, the system can be upgraded by adding advanced features and modern technologies to improve functionality, security, and user experience.

Some possible future enhancements are:

➤ **Integration with Database**

The current system uses file handling for storing inventory data. In the future, it can be integrated with a relational database such as MySQL to improve data management, security, and scalability. A database system would allow better handling of large inventory records and faster data retrieval.

➤ **Graphical User Interface (GUI)**

Currently, the system is console-based. It can be upgraded by developing a user-friendly Graphical User Interface (GUI) using technology such as a web-based framework. This would make the system more interactive, visually appealing, and easier to use.

➤ **Online Inventory Management**

The system can be converted into a web-based application where the cafe owner can monitor stock online. It may also allow multiple users to access the system from different devices within the cafe.

➤ **Automatic Low-Stock Alert System**

An automatic alert feature can be added to notify the user when the quantity of any item falls below a predefined limit. This would help in timely restocking and prevent shortages.

➤ **Barcode Integration**

Future versions of the system can include barcode technology for faster stock entry and updates. Scanning product barcodes would reduce manual data entry and improve accuracy in inventory tracking.

CODE

```
#include <iostream>

#include <fstream>

using namespace std;

class Item{

private:

    int itemID;

    string itemName;

    int quantity;

public:

    void addItem();

    void viewItems();

    void searchItem();

    void updateItem();

    void deleteItem();

};


// Add Item

void Item::addItem(){

    ofstream file("inventory.txt", ios::app);

    cout << "Enter Item ID: ";

    cin >> itemID;

    cout << "Enter Item Name: ";

    cin >> itemName;

    cout << "Enter Quantity: ";
```

```

    cin >> quantity;

    file << itemID << " " << itemName << " " << quantity << endl;

    file.close();

    cout << "Item Added Successfully!\n";

}

// View Items

void Item::viewItems(){

    ifstream file("inventory.txt");

    cout << "\nID\tName\tQuantity\n";

    while (file >> itemID >> itemName >> quantity){

        cout << itemID << "\t" << itemName << "\t" << quantity << endl;

    }

    file.close();

}

// Search Item

void Item::searchItem(){

    int id;

    bool found = false;

    cout << "Enter Item ID to search: ";

    cin >> id;

    ifstream file("inventory.txt");

    while (file >> itemID >> itemName >> quantity) {

        if (itemID == id){

```

```

        cout << "Item Found\n";

        cout << itemID << " " << itemName << " " << quantity << endl;

        found = true;

    }

}

if (!found)

    cout << "Item Not Found\n";

file.close();

}

```

// Update Item

```

void Item::updateItem(){

    int id;

    bool found = false;

    cout << "Enter Item ID to update: ";

    cin >> id;

    ifstream file("inventory.txt");

    ofstream temp("temp.txt");

    while (file >> itemID >> itemName >> quantity){

        if (itemID == id){

            cout << "Enter New Name: ";

            cin >> itemName;

            cout << "Enter New Quantity: ";

            cin >> quantity;

            found = true;

```

```

    }

    temp << itemID << " " << itemName << " " << quantity << endl;

}

file.close();

temp.close();

remove("inventory.txt");

rename("temp.txt", "inventory.txt");

if (found)

    cout << "Item Updated Successfully\n";

else

    cout << "Item Not Found\n";

}

// Delete Item

void Item::deleteItem(){

    int id;

    bool found = false;

    cout << "Enter Item ID to delete: ";

    cin >> id;

    ifstream file("inventory.txt");

    ofstream temp("temp.txt");

    while (file >> itemID >> itemName >> quantity){

        if (itemID != id) {

            temp << itemID << " " << itemName << " " << quantity << endl;

        }

    }

```

```

        else {

            found = true;

        }

    }

file.close();

temp.close();

remove("inventory.txt");

rename("temp.txt", "inventory.txt");

if (found)

    cout << "Item Deleted Successfully\n";

else

    cout << "Item Not Found\n";

}

int main(){

    Item obj;

    int choice;

    do{

        cout << "\n--- Cafe Inventory System ---\n";

        cout << "1. Add Item\n";

        cout << "2. Update Item\n";

        cout << "3. Delete Item\n";

        cout << "4. Search Item\n";

        cout << "5. View Items\n";

        cout << "6. Exit\n";

```



```
        cout << "Enter your choice: ";

        cin >> choice;

if (!(choice >= 1 && choice <= 5))

{

    if (choice == 6)

    {

        cout << "Exiting Program...\n";

    }

    else

    {

        cout << "Invalid Choice\n";

    }

}

else

{

    if (choice == 1)

        obj.addItem();

    else if (choice == 2)

        obj.updateItem();

    else if (choice == 3)

        obj.deleteItem();

    else if (choice == 4)

        obj.searchItem();

    else if (choice == 5)

        obj.viewItems();

}
```

```
}  
}  
while (choice != 6);  
  
    return 0;  
}
```

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